Jane Street April 2025 Puzzle (Sum One, Somewhere)

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1 Introduction

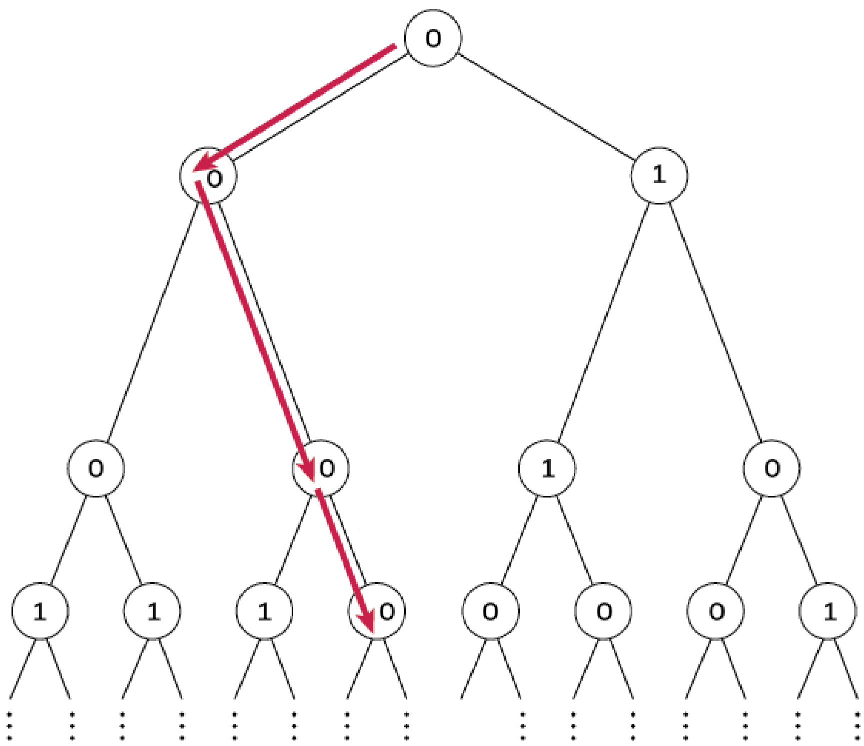

Figure 1: Sketch of the binary tree generated, with the red arrows indicating a possible valid path.

The description of the problem is:

> *"For a fixed p, independently label the nodes of an infinite complete binary tree 0 with probability p, and 1 otherwise. For what p is there exactly a 1/2 probability that there exists an infinite path down the tree that sums to at most 1 (that is, all nodes visited, with the possible exception of one, will be labeled 0). Find this value of p accurate to 10 decimal places."*

2 Setting up the problem

The problem states that there is an infinitely deep binary tree, where each node is labeled '0' with probability p, and '1' with probability $(1-p)$. The focus of the problem is on finding the root-to-leaf paths with at most one "1" along it (equivalently, all but at most one of the nodes on that path are labeled 0). Specifically, we want to compute the probability p such that the chance that a path with at most one '1' exists is $\frac{1}{2}$.

Some insights to solve the problem are:

- The tree is infinitely deep, so any node is statistically equivalent to the root node (self similarity). That means that we can use a recursive relation to find the desired probability.
- Every node is independent from any other node, they are all Bernoulli random variables.
- We note that there are two possible paths that we need to consider, those that are composed entirely of '0's all the way down, and those that have one '1' and then all '0's.

Given the above, let us define two different probabilities, P_0 and P_1. P_0 is the probability that starting at a given node, there exists a path downward that is made entirely of '0's. Similarly, P_1 is the probability that starting at a given node, there exists a path downward with at most one '1' and only '0's otherwise. Of course, we want to find P_1 eventually, which is applicable to the root node.

Since each node is equivalent to its parent, we can set up recursive relationships for P_1 and P_0 purely based on the value at the node.

- Case 1: If the current node is a '0' (happening with probability p. In this case, the current node itself does not use up the allowance of a '1' on the path. We still have the same allowance for the rest of the path (at most one '1' can appear among the descendants). For an infinite acceptable path to exist starting at this node, at least one of its two children must lead into a continuing acceptable path. The left child's subtree and right child's subtree are independent and each is identical in distribution to the whole tree. From the perspective of a child node, we are still allowed at most one '1' on the path (since none has been used so far). The probability that a given child's subtree has an acceptable path is P_1 (by definition). Therefore, the probability that neither child leads to an acceptable path is $(1-P_1) \times (1-P_1) = (1-P_1)^2$. Thus, the probability that at least one child does yield an acceptable path is $1-(1-P_1)^2$. Combining these, if the current node is 0, the probability that an acceptable infinite path exists through this node is $1-(1-P_1)^2$. (We multiply by p later to account for the probability the node is 0 in the first place.)
- Case 2: If the current node is a '1' (happening with probability $(1-p)$. In this case, the current node uses up the single allowed '1' for the path. This means any continuing path downward from here cannot contain any further 1's – all descendants on that path must be 0. In other words, after this node, we switch to the more restrictive scenario of no 1's allowed. For an infinite path to exist starting at this node under these conditions, at least one of the two children must be the start of an all-zeros infinite path. The probability that a given child's subtree has an infinite all-zero path is P_0 (by definition). By similar reasoning as in Case 1, the probability that at least one child has an all-zero path is $1-(1-P_0)^2$. So if the current node is 1, the probability an acceptable path exists through it (with the current node being the one exception) is $1-(1-P_0)^2$. (Again, to be weighted by $1-p$ for the node being 1.)

Now we combine these cases, weighting by the probability of the current node's label. For a given node (in the state where one '1' is allowed up to this point), the probability P_1 satisfies:

$$P_1 = p * [1-(1-P_1)^2] + (1-p) * [1-(1-P_0)^2] \tag{1}$$

Similarly, we can write a self-consistent equation for P_0. If a node is in the state of no 1's allowed (meaning the path from this node down must be all zeros): If the node is labeled 0 (probability p), then we require at least one child's subtree to also have an all-zero infinite path. The probability a child subtree has an all-zero path is P_0, so the chance at least one child succeeds is $1-(1-P_0)^2$. If the node is labeled 1 (probability $1-p$) under the "no-1-allowed" condition, then it is impossible to continue an all-zero path (the path fails immediately because a '1' appears when none are allowed). Therefore, the equation for P_0 is:

$$P_0 = p * [1-(1-P_0)^2] + (1-p) * 0 \tag{2}$$

which is equivalent to:

$$P_0 = p * [1-(1-P_0)^2] \tag{3}$$

These two equations are recursive fixed-point equations for P_0 and P_1, reflecting the self-similar structure of the infinite tree. They capture the probabilities in terms of themselves (and p), which we can solve for consistent values.

3 Solving the recursive fixed-point equations

The next step is to solve these equations for P_0 and P_1 as functions of p. Then we will determine the specific p that makes $P_1 = 1/2$. Here's how to proceed:

- Solve for P_0 first since the P_0 equation is simpler.
  - Algebraically, $P_0 = p[1-(1-P_0)^2]$. Expand the square: $1-(1-P_0)^2 = 1-(1-2P_0+P_0^2) = 2P_0-P_0^2$. So the equation becomes $P_0 = p(2P_0 - P_0^2)$.
  - Rearranged, this is a quadratic in P_0: $p, P_0^2 - 2p, P_0 + P_0 = 0$, or $p, P_0^2 - (2p-1)P_0 = 0$. The solutions are $P_0 = 0$ or $P_0 = \frac{2p-1}{p}$ (assuming $p \neq 0$).
  - Interpretation: $P_0 = 0$ is a valid solution for P_0 when $2p - 1 < 0$, i.e. when $p \leq 1/2$. In fact, if $p \leq 0.5$, the probability of an all-zero infinite path is zero (intuitively, when each node is 0 with probability at most 50%, an infinite chain of 0's almost surely does not exist).
  - For $p > 1/2$, the second solution $P_0 = \frac{2p-1}{p}$ is the relevant one (and lies between 0 and 1). This gives the probability of an all-zero path in the supercritical case where such a path can occur with positive probability. (For example, if $p = 1$ all nodes are 0 and $P_0 = 1$; if p is just above 0.5, P_0 is small but positive.)
  - We will be interested in the regime $p > 1/2$, because for $p \leq 1/2$ even a single acceptable path has probability 0 (which is certainly below 1/2). So going forward, assume $p > 0.5$ and use $P_0 = \frac{2p-1}{p}$.
- Plug P_0 into the P_1 equation and solve for P_1.
  Recall the P_1 equation:

  $$P_1 = p * [1 - (1 - P_1)^2] + (1 - p) * [1 - (1 - P_0)^2] \tag{4}$$

  We can substitute the expression for P_0 (for $p > 0.5$) into this. First, simplify the pieces:

  $$1 - (1 - P_1)^2 = 2P_1 - P_1^2 \tag{5}$$

  $$1 - (1 - P_0)^2 = 2P_0 - P_0^2 \tag{6}$$

  So the equation becomes:

  $$P_1 = p(2P_1 - P_1^2) + (1 - p) * (2P_0 - P_0^2) \tag{7}$$

  This is a nonlinear (quadratic) equation in P_1, but we treat p as a constant for now. It may be helpful to expand and rearrange it into a standard form:

  $$pP_1^2 + (1 - 2p)P_1 - (1 - p)(2P_0 - P_0^2) = 0 \tag{8}$$

  This equation links P_1 and known quantities (p and P_0). In principle, one could solve this quadratic for P_1 in terms of p. However, our primary interest is finding p such that $P_1 = 1/2$. So we can plug $P_1 = 0.5$ directly into this equation, using the earlier expression for P_0:
  We have $P_0 = \frac{2p-1}{p}$ (for $p > 0.5$). Compute $2P_0 - P_0^2$ with this value to simplify the equation. In fact, $2P_0 - P_0^2$ simplifies nicely:

  $$2P_0 - P_0^2 = \frac{2p - 1}{p^2} \tag{9}$$

  Now substitute $P_1 = 0.5$ and $2P_0 - P_0^2 = \frac{2p-1}{p^2}$ into the P_1 equation. Setting $P_1 = 0.5$ should satisfy:

  $$\frac{1}{2} = p(2 * \frac{1}{2} - (\frac{1}{2})^2) + (1 - p)\frac{2p - 1}{p^2} \tag{10}$$

Simplify the left part: $p(2(0.5) - 0.25) = p(1 - 0.25) = p(0.75) = 0.75p$. So the equation becomes:

$$\frac{1}{2} = \frac{3}{4}p + (1-p)\frac{2p-1}{p^2} \tag{11}$$

This is an equation purely in p. (All the P_1 and P_0 terms are eliminated by setting the condition.)

- Solve for p.
  Now we solve the above equation for p. It might be easiest to bring everything to one side and find a common denominator to solve it as a polynomial equation. Multiplying through by p^2 to clear the fraction, we get:

  $$\frac{1}{2}p^2 = \frac{3}{4}p^3 + (1-p)(2p-1) \tag{12}$$

  After some algebra tricks, we arrive at the final equation for p:

  $$3p^3 - 10p^2 + 12p - 4 = 0 \tag{13}$$

  We can then use the formulas for solution of cubic equations to find the value of p. Ignoring the solutions with imaginary coefficients, we determine that:

  $$p = \frac{1}{9}\left(10 - \frac{4 * 2^{2/3}}{\sqrt[3]{9\sqrt{57}-67}} + \sqrt[3]{2(9\sqrt{57}-67)}\right) \approx 0.53060357543... \tag{14}$$

The result $p \approx 0.53$, lies just above 0.5, reflecting the intuitive threshold at which an infinite binary tree transitions from being too sparse to support even a single acceptable path, to being just dense enough for such a path to emerge with meaningful probability, highlighting the delicate balance between randomness and structure in infinite systems.